Phonological Competition in Hebrew Visual Word Recognition: Evidence for Inner Speech from Homophone Processing

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When readers silently process written words, do they "hear" those words in their heads? Evidence suggests that inner speech, an internalized phonological representation activated without overt articulation, plays a central role in silent reading (Perrone-Bertolotti et al., 2012). However, less is known about how different sources of ambiguity, phonological versus orthographic, differentially engage competition during silent reading in a language with an underspecified orthography. Hebrew's unvoweled orthography provides a unique test case: readers must internally generate phonological forms from consonantal input, making phonological processing unavoidable (Frost, 2012). While phonological activation during Hebrew word recognition has been demonstrated (Bitan et al., 2017), it remains unknown how inner speech phonology interacts with various types of lexical ambiguity when multiple phonological and semantic representations compete.

We examined how phonological activation through inner speech affects the processing of lexical ambiguity during silent reading. 49 native Hebrew speakers (ages 19-28) completed a single-word lexical decision task, processing three types of ambiguous words along with frequency and length matched unambiguous controls: homophones (same phonology, different orthography and meaning; the less frequent orthographic form was presented to maximize the possibility for activation of both forms; e.g., עף/אף *af*, ”nose”/”flew”), homographs (same orthography, different phonology and meaning; e.g., ספר *sefer/sapar*, ”book”/”hairdresser”), and homonyms (same orthography and phonology, different meanings; e.g., מפה *mapa*, ”tablecloth”/”map”). Participants judged whether letter strings were real Hebrew words, pressing designated keys for word/non-word responses. Each ambiguous condition included 21 items, along with 63 unambiguous words and 126 non-words as fillers (252 items total, presented in random order). Response times (RTs) and accuracy were analyzed using linear mixed-effects models with maximal random-effects structure for participants and items.

Results revealed a notable selective ambiguity effect. Homophones incurred the largest processing cost, with significantly slower RTs ($M = 815$ms vs. 647ms; $\beta = 172.10$, $p < .001$) and lower accuracy (85% vs. 98%; $\beta = -0.13$, $p = .003$) compared to unambiguous controls. This suggests that when orthography differs but phonology aligns in inner speech, multiple lexical candidates compete for selection, creating substantial processing demands. Homographs showed numerically slower responses ($M = 663$ms vs. 631ms; $\beta = 34.78$, $p = .21$), though this difference was not statistically significant. Homonyms showed no processing difference from controls ($M = 635$ms vs. 641ms; $\beta = -5.18$, $p = .81$). Homographs (99% vs. 98%; $\beta = 0.009$, $p = .47$) and homonyms (98% vs. 99%; $\beta = -0.006$, $p = .60$) did not differ significantly from their matched unambiguous controls in accuracy. Although ambiguous words were slower overall than controls ($M = 703$ms vs. 640ms; $\beta = 67.02$, $p = .001$), this effect appears to be driven primarily by homophones.

This study provides a systematic comparison of how different types of ambiguity influence silent word recognition in Hebrew. The pronounced homophone effect demonstrates that inner speech creates sufficiently specified phonological representations to activate orthographically distinct lexical entries, leading to competition at the phonological-orthographic interface. This indicates that phonological similarity increases lexical competition even when orthographic forms differ, consistent with the activation of speech-like phonological codes during silent reading. Results support interactive models of reading (Harm & Seidenberg, 2004) in which orthographic input automatically triggers phonological activation. This provides a key behavioral baseline for neuroimaging studies aimed at decoding inner speech during natural reading.

References

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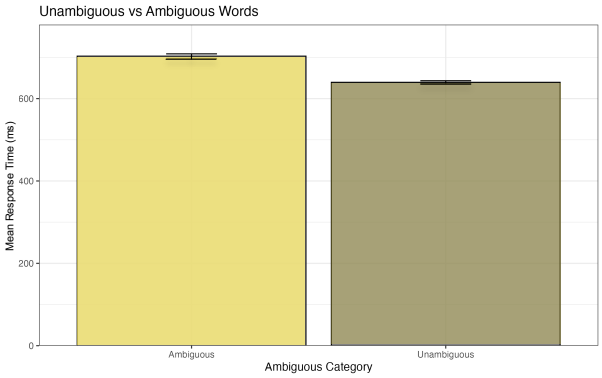

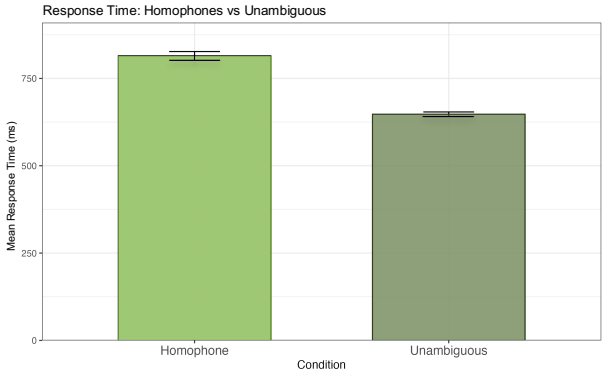

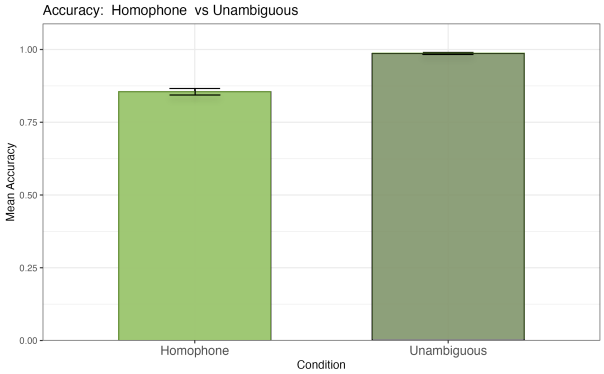

Example Stimuli

Condition	Hebrew Stimulus	Phonetic Transcription	Meaning
Homophone	עף	*af*	flew (עף) / nose (אף)
Homophone	סערה	*se'ara*	storm (סערה) / hair (שערה)
Homonym	מפה	*mapa*	map / tablecloth
Homonym	כדור	*kadur*	ball / pill
Homograph	ספר	*sefer / sapar*	book / hairdresser
Homograph	מדבר	*midbar / medaber*	desert / speaking

Homophones presented in less frequent orthographic form. Controls were frequency- and length-matched unambiguous words. Non-words were phonotactically legal (e.g., דז *dez*, טשל *tashel*).